

# Cyril Shih-Huan Hsu

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## Professional Summary

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**AI researcher with a PhD in Informatics**, specializing in deep reinforcement learning, Transformer-based architectures, generative models, and multi-agent systems. Research focuses on decision-making, optimization, meta-learning, and coordination in complex networked environments, with applications spanning large-scale systems and decentralized agentic systems. Published in top-tier IEEE Transactions and leading international conferences.

## Education

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| <b>PhD's degree in Informatics</b><br><i>Doctor of Science</i>                                  | <i>University of Amsterdam</i><br>2021.7 - 2025.7    |
| <b>Master's degree in Electrical Engineering</b><br><i>Master of Science</i>                    | <i>National Taiwan University</i><br>2013.9 - 2015.6 |
| <b>Bachelor's degree in Bio-environmental Systems Engineering</b><br><i>Bachelor of Science</i> | <i>National Taiwan University</i><br>2009.9 - 2013.6 |

## Work experience

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| <b>University of Amsterdam</b><br><i>Postdoctoral Researcher</i>                                                                                                                                                                                                                                                                                                       | 2025.9 - present<br><i>Amsterdam, The Netherlands</i> |
| <ul style="list-style-type: none"><li>• Agentic AI-Driven Network Orchestration</li></ul>                                                                                                                                                                                                                                                                              |                                                       |
| <b>University of Amsterdam</b><br><i>Teaching Assistant/Student Thesis Supervisor</i>                                                                                                                                                                                                                                                                                  | 2021.7 - 2025.7<br><i>Amsterdam, The Netherlands</i>  |
| <ul style="list-style-type: none"><li>• Advanced Deep Learning (Course)</li><li>• Advanced Networking (Course)</li><li>• Resource Optimization for Facial Recognition Systems (Thesis)</li><li>• Multi-domain Simulation Framework for Virtualized Infrastructures (Thesis)</li><li>• ML-driven Object Detection Task Offloading for Edge Computing (Thesis)</li></ul> |                                                       |
| <b>NEC Laboratories Europe</b><br><i>Visiting Researcher</i>                                                                                                                                                                                                                                                                                                           | 2024.7 - 2024.12<br><i>Heidelberg, Germany</i>        |
| <ul style="list-style-type: none"><li>• Vision Transformer-Based Framework for Real-Time Radio Quality Prediction</li><li>• Distributed Vision-Language Model-Based Multi-Robot Coordination System</li></ul>                                                                                                                                                          |                                                       |
| <b>Deep Cube SA</b><br><i>CTO &amp; Co-founder</i>                                                                                                                                                                                                                                                                                                                     | 2018.8 - 2021.5<br><i>Lausanne, Switzerland</i>       |
| <ul style="list-style-type: none"><li>• Glaucoma / Skin Disorder Screening System</li><li>• Data Collection Platform for Medical Centers</li></ul>                                                                                                                                                                                                                     |                                                       |
| <b>Relajet Technologies Ltd.</b><br><i>Technical Director of Artificial Intelligence</i>                                                                                                                                                                                                                                                                               | 2018.2 - 2020.2<br><i>Taipei, Taiwan</i>              |
| <ul style="list-style-type: none"><li>• Real-time Voice Identification/Speech Enhancement System</li></ul>                                                                                                                                                                                                                                                             |                                                       |

- Real-time Facial Recognition System
- Multimodal Interactive Chatbot
- Deep Neural Networks for Embedded System

### *Language proficiencies*

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| <b>English</b>  | Fluent |
| <b>Mandarin</b> | Native |

### *Publication List*

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| <b>Journal</b>    | <b>Hsu, C. S. H.</b> , Dalgkitis, A., Papagianni, C., & Grosso, P. (2026). Transformer-Empowered Actor-Critic Reinforcement Learning for Sequence-Aware Service Function Chain Partitioning. In 2026 IEEE Transactions on Network Science and Engineering (TNSE).                                                                    |
| <b>Journal</b>    | <b>Hsu, C. S. H.</b> , Martín-Pérez, J., De Vleeschauwer, D., Valcarenghi, L., Li, X., & Papagianni, C. (2025). A Deep RL Approach on Task Placement and Scaling of Edge Resources for Cellular Vehicle-to-Network Service Provisioning. In 2025 IEEE Transactions on Network and Service Management (TNSM).                         |
| <b>Journal</b>    | <b>Hsu, C. S. H.</b> , De Vleeschauwer, D., & Papagianni, C. (2023). SLA Decomposition for Network Slicing: A Deep Neural Network Approach. In 2023 IEEE Networking Letters (NL).                                                                                                                                                    |
| <b>Conference</b> | <b>Hsu, C. S. H.</b> , Cheng, W. Y. C., & Papagianni, C. (2026). Progressive Semantic Communication for Efficient Edge-Cloud Vision-Language Models. In 2026 IEEE Global Communications Conference (GLOBECOM).                                                                                                                       |
| <b>Conference</b> | <b>Hsu, C. S. H.</b> , Li, X., Zanzi, L., Yang, Z., Papagianni, C., & Costa-Perez, X. (2026). MapViT: A Two-Stage Vision Transformer-Based Framework for Real-Time Radio Quality Map Prediction in Dynamic Environments. In 2026 IEEE International Conference on Communications (ICC).                                              |
| <b>Conference</b> | Dalgkitis, A., <b>Hsu, C. S. H.</b> , Papagianni, C., Grosso, P. (2026). LLM-ASF6G: LLM-Based Algorithm Selection Framework for 6G Network Service Optimization. In 2026 Joint European Conference on Networks and Communications & 6G Summit (EuCNC).                                                                               |
| <b>Conference</b> | Dalgkitis, A., <b>Hsu, C. S. H.</b> , Papagianni, C., Grosso, P., & de Laat, C. (2025). LLM-based Optimization Algorithm Selection for High-Performance Networks Orchestration. Proceedings of the SC '25 Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis (SC Workshops). |
| <b>Conference</b> | <b>Hsu, C. S. H.</b> , Papagianni, C., & Grosso, P. (2025). RAILS: Risk-Aware Iterated Local Search for Joint SLA Decomposition and Service Provider Management in Multi-Domain Networks. In 2025 IEEE Conference on High Performance Switching and Routing (HPSR).                                                                  |
| <b>Conference</b> | <b>Hsu, C. S. H.</b> , Martín-Pérez, J., Papagianni, C., & Grosso, P. (2023). V2N Service Scaling with Deep Reinforcement Learning. In 2023 IEEE Conference on Network Operations and Management Symposium (NOMS).                                                                                                                   |
| <b>Conference</b> | Sun, M. C., <b>Hsu, C. S. H.</b> , Yang, M. C., & Chien, J. H. (2018). Context-Aware Cascade Attention-Based RNN for Video Emotion Recognition. In 2018 First Asian Conference on Affective Computing and Intelligent Interaction (ACII Asia).                                                                                       |
| <b>Conference</b> | Chang, W. Y., <b>Hsu, C. S. H.</b> , & Chien, J. H. (2017). FATAUVA-Net: An Integrated Deep Learning Framework for Facial Attribute Recognition, Action Unit Detection, and Valence-Arousal Estimation. In 2017 IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW).                                        |
| <b>Conference</b> | <b>Hsu, C. S. H.</b> , & Yu, T. L. (2015). Optimization by Pairwise Linkage Detection, Incremental Linkage Set, and Restricted/Back mixing: DSMGA-II. In Proceedings of the 2015 Annual Conference on Genetic and Evolutionary Computation (GECCO).                                                                                  |